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A Framework for Joint Beam Scheduling and Resource Allocation in Beam-Hopping-Based Satellite Systems

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Abstract

With the rapid development of heterogeneous satellite networks integrating geostationary earth orbit (GEO) and low earth orbit (LEO) satellite systems, along with the significant growth in the number of satellite users, it is essential to consider frequency compatibility and coexistence between GEO and LEO systems, as well as to design effective system resource allocation strategies to achieve efficient utilization of system resources. However, existing beam-hopping (BH) resource allocation algorithms in LEO systems primarily focus on beam scheduling within a single time slot, lacking unified beam management across the entire BH cycle, resulting in low beam-resource utilization. Moreover, existing algorithms often employ iterative optimization across multiple resource dimensions, leading to high computational complexity and imposing stringent requirements on satellite on-board processing capabilities. In this paper, we propose a BH-based beam scheduling and resource allocation framework. The proposed framework first employs geographic isolation to protect the GEO system from the interference of the LEO system and subsequently optimizes beam partitioning over the entire BH cycle, time-slot beam scheduling, and frequency and power resource allocation for users within the LEO system. The proposed scheme achieves frequency coexistence between the GEO and LEO satellite systems and performs joint optimization of system resources across four dimensions—time, space, frequency, and power—with reduced complexity and a progressive optimization framework. Simulation results demonstrate that the proposed framework achieves effective suppression of both intra-system and inter-system interference via geographic isolation, while enabling globally efficient and dynamic beam scheduling across the entire BH cycle. Furthermore, by integrating the user-level frequency and power allocation algorithm, the scheme significantly enhances the total system throughput. The proposed progressive optimization framework offers a promising direction for achieving globally optimal and computationally tractable resource management in future satellite networks.

Keywords: beam-hopping; heterogeneous satellite networks; resource allocation; spectrum sharing



Academic Editor: Stefano Scanzio

Received: 14 June 2025

Revised: 14 July 2025

Accepted: 17 July 2025

Published: 18 July 2025

Citation: Zhang, J.; Li, W.; Li, Y.; Wang, H.; Li, S. A Framework for Joint Beam Scheduling and Resource Allocation in Beam-Hopping-Based Satellite Systems. *Electronics* **2025**, *14*, 2887. <https://doi.org/10.3390/electronics14142887>

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1. Introduction

1.1. Motivation

With the large-scale deployment of low earth orbit (LEO) satellite constellations, heterogeneous satellite networks are emerging as a key architectural paradigm for next-generation space–air communications. Reference [1] provides a detailed overview of their

advantages, such as global coverage, low latency, and high capacity. It also highlights key technical challenges, including cross-orbit coordination, interference management, and efficient resource scheduling across LEO and geostationary earth orbit (GEO) satellite layers. These heterogeneous satellite networks combine LEO and GEO satellite systems to exploit their complementary strengths. LEO satellites enable high data rates and low communication latency, while GEO satellites deliver reliable coverage in a wide area [2]. However, this heterogeneous integration also introduces new technical challenges. In particular, the coexistence of LEO and GEO systems operating within overlapping frequency bands can lead to severe inter-system interference, making spectrum sharing a critical issue to be addressed [3]. In addition, to fully exploit the capacity of the system, satellite networks are required to allocate communication resources, such as frequency bands, transmission power, time slots, and beam directions in a highly adaptive and dynamic way. This requirement is particularly critical for LEO satellites, which have greater flexibility in resource scheduling [4]. However, interdependence among these resources across the temporal, spatial, spectral, and power domains results in strong multidimensional coupling, substantially increasing the complexity of joint resource optimization [5].

In conventional multi-beam satellite communication systems, each beam typically provides fixed coverage over a specific geographic region, lacking the flexibility to adapt resource allocation in response to dynamic user distribution. This often leads to resource shortages in high-traffic areas and resource underutilization in low-demand regions, resulting in inefficiencies and waste in the overall allocation of resources of the system [6]. Beam-hopping (BH) technology offers a promising solution to the resource allocation problem characterized by time–space–frequency coupling in heterogeneous satellite networks. By dividing operational time into discrete slots, the BH algorithm enables satellites to dynamically activate beams across different regions based on user demand and channel conditions, thus achieving multiplexed time-divided coverage and adapting to spatio-temporal variations in traffic load [7].

BH technology facilitates multi-dimensional resource allocation across the temporal, spatial, spectral, and power domains. Through this flexible and adaptive resource management paradigm, the system can dynamically schedule and optimize resources, ensuring rapid responsiveness to non-uniform traffic distributions and maintaining service quality guarantees [8]. Furthermore, by leveraging dynamic beam steering capabilities, BH technology enables precise control over spatial coverage, which not only increases the number of users that can be served simultaneously but also improves the efficiency of spatial resource utilization [9]. In addition, BH technology supports dynamic frequency and power allocation across beams, allowing the system to flexibly adapt to diverse user-traffic demands and service requirements [10].

1.2. Related Work

BH has emerged as a promising technique for enhancing the flexibility and efficiency of satellite communication systems. Its performance benefits have been extensively validated in the literature. Specifically, [11] quantifies the throughput improvement of BH over fixed multi-beam illumination at the link-level, while [12] further substantiates these gains at the system-level by explicitly accounting for inter-beam interference. Building upon these foundational studies, [13] characterizes BH-based resource management as a multidimensional optimization problem involving joint time-slot scheduling and frequency–power allocation.

In the domain of beam scheduling, with the growing density of satellite constellations, especially in LEO, the simultaneous coverage of overlapping ground areas by multiple satellites from different orbits has introduced new challenges for beam coordination and

interference mitigation [14]. To support dynamic and uneven traffic demands while minimizing inter-beam interference within a satellite system, coordinated beam steering and resource scheduling have attracted growing research interest [15]. For instance, [16] proposes a two-stage approach that jointly minimizes the number of active beams and balances user load under a fixed power constraint, using graph-theoretic beam clustering and K-means-based user grouping. Similarly, [17] formulates beam placement as a multi-objective optimization problem and introduces a genetic algorithm to obtain Pareto-efficient trade-offs between beam count and spectrum usage. In [18], the authors further extend this line of work by jointly optimizing beam locations and transmitting power using both iterative and deep learning-based solutions, significantly improving computational efficiency.

In parallel, research on frequency and power allocation has progressed along a complementary track. The authors of [19] treat spectrum assignment and power control as separate optimization subproblems. In [20], the authors propose a greedy algorithm for frequency and power allocation in multi-beam LEO systems. The authors of [21] introduce an automatic power control (APC) mechanism to mitigate co-linear interference in the uplink and downlink between LEO and GEO systems, thereby effectively reducing inter-system interference. The authors of [2] presented a joint beam management and power allocation framework focused on maximizing GEO system throughput while ensuring continuity for LEO links. Furthermore, [22] presents a frequency and power allocation strategy tailored for multi-beam systems, designed to meet non-uniform user distribution and traffic demands under constrained power budgets, while exploring the trade-off between total system capacity and user fairness. In [23], the authors optimize subchannel and power allocation in a multi-beam satellite communication system to meet traffic requirements, while simultaneously minimizing power consumption and bandwidth overhead.

1.3. Contribution

Despite the advancements in the existing literature, two critical challenges remain inadequately addressed. Firstly, existing BH technology resource allocation methods primarily focus on beam assignment within a single time slot, lacking global optimization across the entire BH cycle. This limits the effective utilization of temporal resources within the system. Secondly, while these studies have provided important theoretical and algorithmic foundations, most either target a single resource dimension or rely on fully coupled optimization models that impose high computational complexity. Resource allocation in BH-based systems involves multiple coupled dimensions—beam-footprint partitioning, time-slot scheduling, and frequency–power allocation. The resulting joint optimization problem is high-dimensional and computationally complex. Therefore, it is essential to develop a scheduling framework that can effectively decouple spatio-temporal-spectral resources and adopt a hierarchical optimization strategy to achieve near-optimal system performance with reduced computational complexity.

To address the challenges of high-dimensional coupling and suboptimal temporal utilization in BH-based satellite systems, this paper investigates a BH-based beam scheduling and resource allocation framework. In contrast to conventional approaches that rely on joint iterative optimization, the proposed framework adopts a progressive, dimension-wise optimization strategy, in which each resource dimension is handled independently in a sequential manner, without requiring cross-iteration between modules. This enables efficient and scalable scheduling with significantly reduced computational complexity. Specifically, we develop a spatio-temporal-frequency resource allocation framework tailored for heterogeneous GEO–LEO satellite networks. The objective is to fully exploit the time, spatial, and spectral resources available to the LEO system, thereby enhancing the overall system throughput. Within the proposed framework, beam-footprint partitioning,

time-slot assignment, beam-level power allocation, and user-specific frequency–power allocation are jointly addressed over the entire BH cycle. Each component is independently modeled and solved using modular algorithms, including clustering and integer linear programming (ILP) for spatial optimization, a hybrid greedy–simulated annealing approach for temporal scheduling, and a Hungarian–water-filling–convex optimization pipeline for spectral and power domains. Although the resource dimensions are optimized sequentially, cross-domain dependencies are minimized and there is no need for iterative feedback between stages, thereby enabling tractable and flexible resource coordination under realistic constraints.

The main contributions of this paper are summarized as follows:

- **Beam-Footprint Partitioning:** to improve the coverage efficiency of multi-beam LEO satellite systems, we develop a multi-phase beam partitioning strategy that integrates density-aware clustering with ILP. The initial beam layout is guided by user distribution, where regions of high user density are prioritized during clustering. Subsequently, ILP is applied to optimize the association between users and beams, aiming to adjust beam center positions so that users are located as close as possible to beam centers for enhanced channel gain. An iterative adjustment step eliminates low-efficiency beams and reallocates their users, thereby increasing beam utilization efficiency and enabling better adaptation to spatially non-uniform traffic patterns.
- **Time-Slot Beam Scheduling:** for scheduling beam positions across time slots within a BH cycle, we develop a hybrid approach, combining a greedy algorithm and simulated annealing. A feasible initial schedule is generated using a greedy filling strategy under constraints on minimum beam separation and per-slot beam limits. A simulated annealing algorithm is then applied to explore the solution space through probabilistic perturbations, enhancing the global optimality of beam allocation.
- **Frequency–Power Joint Allocation:** we propose a joint optimization method that integrates the Hungarian algorithm and water-filling theorem. The Hungarian algorithm is used to efficiently match users with sub-bands, while water-filling is applied to optimize intra-beam power distribution. Finally, we apply convex optimization to regulate inter-beam power allocation, leading to enhanced system throughput for the LEO network.

The remainder of this paper is structured as follows. Section 2 introduces the system model and formulates the joint resource allocation problem in the heterogeneous GEO–LEO satellite network, encompassing beam-footprint partitioning, time-slot beam scheduling, and frequency–power allocation with the goal of maximizing the overall system throughput. Section 3 details the proposed beam-hopping-based resource allocation framework, which employs a progressive, dimension-wise optimization strategy. Section 4 presents numerical simulations and analyzes the performance of the proposed algorithms in terms of throughput and computational efficiency. Finally, Section 5 summarizes the key findings and outlines potential directions for future research.

2. System Model

2.1. System Setup

This study investigates the resource scheduling problem within a single BH cycle, in line with the scheduling characteristics of BH systems. Based on the analysis in [24], the ground-track displacement of a LEO satellite within a single BH cycle is negligible compared to the beam coverage radius, and user positions are assumed to be static over this short duration. Thus, we adopt a snapshot-based model where satellite positions and user distribution remain unchanged within each BH cycle.

The two satellite systems operate over a shared frequency spectrum but serve distinct service objectives: the LEO network is designed for dynamic, high-density user access, whereas the GEO network is tasked with providing stable and continuous coverage. In the simulation scenario, a representative ground service area is selected, which is served by a heterogeneous satellite network consisting of N_{sat} LEO satellites and one GEO satellite as shown in Figure 1.

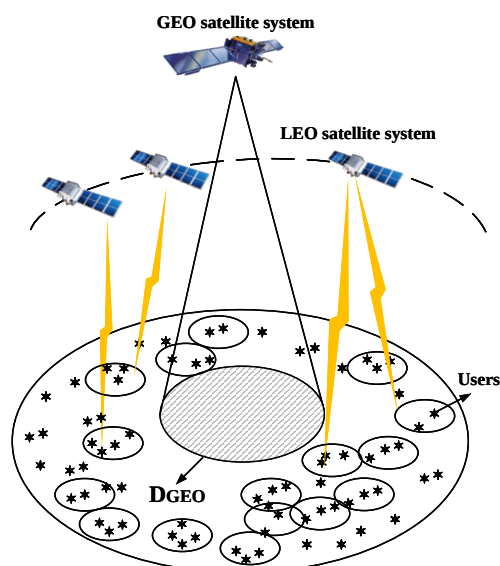


Figure 1. GEO-LEO scenario schematic diagram.

In the time domain, one BH cycle is divided into M discrete time slots. During each time slot, each satellite determines the positions of its activated beams. In the spatial domain, each LEO satellite is equipped with a multi-beam antenna array and can simultaneously illuminate up to N_{beam} beams, with each beam covering a circular area of radius R . Within a single BH time slot, the total number of active beams across all N_{sat} satellites is constrained by $N = N_{\text{sat}} \cdot N_{\text{beam}}$. To mitigate intra-system interference, the illuminated beams in each time slot must maintain a minimum inter-beam distance $d_{\text{min}} \geq 4R$ as suggested in [25]. To mitigate inter-system interference, a protection region D_{GEO} is established around the GEO users, within which the activation of LEO satellite beams is prohibited. In the remainder of this study, all LEO users are assumed to be located outside of the GEO protection region to ensure inter-system interference is avoided during resource scheduling [26]. In the frequency domain, each LEO satellite is allocated the entire system bandwidth B_s , and each beam can serve up to L users, with the bandwidth equally distributed among them. The total transmit power of each satellite is limited to P_s and the sum of the power allocated to all beams must not exceed this limit.

By partitioning the time, space, and frequency domains, a three-dimensional spatio-temporal-frequency resource grid is constructed. Within a single BH cycle, the N_{sat} LEO satellites can schedule up to MNL resource units, where each user is limited to occupying at most, one unit of this grid. To represent user-resource assignment, we define a binary variable $\alpha_{t,b,f}^i \in \{0, 1\}$, which indicates whether the i -th user is assigned to the t -th time slot, the b -th beam, and the f -th sub-band. For a given user i , the received signal power can be expressed as follows:

$$S_{t,b,f}^i = P_{t,b,f}^i \cdot G_b^i \cdot PL_{b,f}^i \quad (1)$$

where $P_{t,b,f}^i$ denotes the transmit power allocated to the i -th user on the f -th sub-band by the b -th beam in the t -th time slot. G_b^i represents the beamforming gain from the b -th beam

to the i -th user, and $PL_{b,f}^i$ denotes the path loss between the b -th beam and the i -th user. The receiver antenna at the user terminal is assumed to be omnidirectional, and thus its gain is omitted.

The interference received by the i -th user is defined as follows:

$$I_{t,b,f}^i = \sum_{b' \neq b} P_{t,b',f}^i \cdot G_{b'}^i \cdot PL_{b',f}^i \quad (2)$$

where $b' \neq b$ represents the aggregated interference from non-serving beams associated with irrelevant links.

Accordingly, the signal-to-interference-plus-noise ratio (SINR) γ^i of the i -th user can be expressed as follows:

$$\gamma_{t,b,f}^i = \frac{P_{t,b,f}^i \cdot G_b^i \cdot PL_{b,f}^i}{\sum_{b' \neq b} P_{t,b',f}^i \cdot G_{b'}^i \cdot PL_{b',f}^i + N_i} \quad (3)$$

where N_i denotes the thermal noise power associated with the sub-band allocated to the i -th user, which is determined by the sub-band bandwidth and receiver characteristics.

The achievable data rate $R_{t,b,f}^i$ for the i -th user can be calculated based on the Shannon capacity formula:

$$R_{t,b,f}^i = B_{t,b,f}^i \log_2(1 + \gamma^i) \quad (4)$$

where $B_{t,b,f}^i$ is the bandwidth of the sub-band assigned to the i -th user via the b -th beam.

2.2. Channel Model

Considering realistic satellite-to-ground propagation conditions, the path loss of the i -th user when served by the b -th beam on the f -th sub-band is composed of three main components: free-space attenuation, atmospheric losses, and Rician fading. The overall path loss is expressed as follows [27]:

$$PL_{b,f}^i = \left(\frac{c}{4\pi df} \right)^2 A^{-1}(d) \rho_1 \quad (5)$$

where ρ_1 is the Rician fading factor, d denotes the distance between the b -th beam and the i -th user, and $A(d)$ represents atmospheric attenuation, which can be further expressed as follows: $A(d) = 10^{\frac{d(4.343\rho_2 + \rho_3)}{10H_S}}$, where ρ_2 and ρ_3 are the cloud and rain attenuation coefficients, respectively. H_S is the orbital altitude of the satellite.

2.3. Problem Formulation

Within a single BH cycle, the N_{sat} LEO satellites can schedule up to MNL resource units and each user is allowed to occupy at most, one such unit. The overall optimization objective is to maximize the utilization of these multi-dimensional resources while jointly optimizing the allocation of frequency and power for each satellite's beams and associated users, so as to maximize the total system throughput. However, due to the strong coupling across the spatio-temporal-frequency-power dimensions, direct optimization of the global objective incurs prohibitively high computational complexity. To address this, we decompose the original problem into two sub-objectives, enabling a stepwise optimization approach. This hierarchical strategy allows for the design of low-complexity algorithms capable of achieving near-optimal solutions with significantly reduced computational overhead.

The first optimization objective aims to maximize the number of scheduled system resources, which can be formulated as the following objective function, P1:

$$\text{P1: } \max \sum_{t=1}^M \sum_{b=1}^N \sum_{f=1}^L \alpha_{t,b,f}^i \quad (6a)$$

$$\text{s.t. } \sum_i \alpha_{t,b,f}^i \leq 1, \forall \text{triplet}\{t, b, f\} \quad (6b)$$

$$\sum_t \sum_b \sum_f \alpha_{t,b,f}^i \leq 1, \forall i \quad (6c)$$

$$\alpha_{t,b,f}^i \in \{0, 1\}. \quad (6d)$$

In objective function P1, constraint (6b) ensures that each spatio-temporal-frequency resource unit can be assigned to at most, one user within a single BH cycle. Constraint (6c) guarantees that each user can be allocated no more than one such resource unit per BH cycle. Constraint (6d) specifies that $\alpha_{t,b,f}^i$ is a binary allocation variable.

The second optimization objective aims to allocate satellite beam power, intra-beam user power, and user frequency resources in order to maximize the total throughput of users served by the satellite system. This objective can be formulated as the following optimization problem, P2:

$$\text{P2: } \max \sum_{t=1}^M \sum_{b=1}^N \sum_{i=1}^{U_{t,b}} R_{t,b,f_i}^i \quad (7a)$$

$$\text{s.t. } \sum_{i=1}^{U_{t,b}} P_t^i \leq P_t^b, \forall t, b \quad (7b)$$

$$P_t^i \geq 0, \forall i, t \quad (7c)$$

$$P_t^b \leq P_{\max}, \forall t, b \quad (7d)$$

$$\sum_{b=1}^N P_t^b \leq P_s, \forall t \quad (7e)$$

$$\sum_{i=1}^{U_{t,b}} f_{t,b}^i = B_s, \forall t, b \quad (7f)$$

$$f_{t,b}^i \geq 0, \forall i, t \quad (7g)$$

$$i \notin D_{\text{GEO}}. \quad (7h)$$

$U_{t,b}$ denotes the number of users served by the b -th beam during the t -th time slot and $\{u_{t,b}^1, u_{t,b}^2, \dots, u_{t,b}^{U_{t,b}}\}$ represents the corresponding user set.

In objective function P2, the associated constraints are interpreted as follows:

- Constraint (7b) specifies that the total power allocated to users under each beam must not exceed the power assigned to that beam.
- Constraint (7c) specifies that the power allocated to each user must be non-negative.
- Constraint (7d) specifies that the power allocated to each beam must not exceed the predefined beam-level power limit.
- Constraint (7e) specifies that the total power allocated across all beams must not exceed the satellite's onboard power budget.
- Constraint (7f) specifies that the total bandwidth allocated to users under each beam must equal the total bandwidth available to the satellite.
- Constraint (7g) specifies that the bandwidth allocated to each user must be non-negative.
- Constraint (7h) specifies that scheduled LEO users must be located outside the GEO protection zones in order to eliminate potential interference to GEO system operations.

Based on the two objective functions defined above, we perform dimension-wise optimization for spatio-temporal-frequency-power resource allocation in the LEO system.

3. A Beam-Hopping-Based Global Beam Scheduling and User Resource Allocation Algorithm

In this section, we propose a beam-hopping-based joint beam scheduling and resource allocation framework for a resource-constrained heterogeneous GEO-LEO satellite network. The proposed framework consists of three core components: beam-footprint partitioning, time-slot-based beam scheduling, and joint frequency-power allocation. The first objective function P1 is addressed through a two-step process involving beam partitioning and beam scheduling, with the goal of maximizing the number of scheduled spatio-temporal-frequency resources. The second objective function P2 is solved via frequency and power allocation, aiming to maximize the total system throughput.

Section 3.1 presents the optimization of beam-footprint partitioning using an ILP approach. Section 3.2, we address the scheduling of beam activations across time slots by proposing a global beam-slot allocation strategy that accounts for spatio-temporal dynamics. Section 3.3 introduces a joint optimization of beam-level power allocation and user-level frequency and power assignment.

3.1. Algorithm A: Beam-Footprint Partitioning Algorithm

In this section, we propose an ILP-based beam-footprint partitioning algorithm. The Algorithm A predefines the set of beams that can be scheduled for the entire BH cycle of the LEO system, based on the spatial distribution of LEO users located outside the GEO protection zones. The objective is to maximize the number of users served per beam, ensuring that each beam is utilized as fully as possible.

3.1.1. Initial Beam Candidate Generation

In the initial beam candidate generation phase, a density-weighted clustering strategy is applied to account for the spatial distribution of users. For each user, the number of neighboring users within a radius R is computed as a local density indicator. Users in high-density regions are selected and replicated to form a virtual user set, thereby guiding the K-means algorithm to place more cluster centers in areas with concentrated user demand. Clustering this density-enhanced user set yields a preliminary set of beam center coordinates, denoted as $\{\mu_1, \mu_2, \dots, \mu_{MN}\}$, by minimizing the sum of squared distances between users and their assigned cluster centers:

$$\min \sum_{j=1}^{MN} \sum_{x_i \in C_j} \|x_i - \mu_j\|^2 \quad (8)$$

where x_i denotes the coordinate of i -th user, and C_j represents the set of user coordinates covered by the j -th beam.

3.1.2. Beam-User Matching

After obtaining the initial beam center coordinates, an optimization model is constructed to maximize the number of users served by each beam. A binary decision matrix of size $U \times MN$ is defined, where U denotes the number of users and MN denotes the number of candidate beams. Each element in this matrix is a binary decision variable denoted as x_{ij} , where $i \in \{1, 2, \dots, U\}$ and $j \in \{1, 2, \dots, MN\}$. If $x_{ij} = 1$, it indicates that the i -th user is scheduled by the j -th beam; conversely, if $x_{ij} = 0$, the i -th user is not served by the j -th beam. The total number of successfully scheduled users can be computed by summing all

elements across both user and beam dimensions of the decision matrix. Accordingly, the optimization objective can be expressed as follows:

$$\max \sum_{i=1}^U \sum_{j=1}^{MN} x_{ij} \quad (9a)$$

$$\text{s.t. } x_{ij} \cdot d_{ij} \leq R, \forall i, j \quad (9b)$$

$$\sum_{i=1}^U x_{ij} \leq F, \forall j \quad (9c)$$

$$\sum_{j=1}^{MN} x_{ij} \leq 1, \forall i. \quad (9d)$$

where d_{ij} denotes the distance between the i -th user and the center of the j -th beam. The constraints are interpreted as follows: constraint (9b) represents the coverage constraint, where a user is eligible for scheduling by a beam only if it is located within the beam's coverage radius R . If $d_{ij} > R$, then x_{ij} must be 0. Constraint (9c) is a capacity constraint, which states that each beam can serve no more than F users. Constraint (9d) is also a capacity constraint, ensuring that each user can be scheduled by at most, one beam.

3.1.3. Iterative Optimization

Due to the possibility that the algorithms in step 1 and step 2 may generate beams with insufficient user coverage, an iterative refinement mechanism is introduced to improve overall beam utilization. Specifically, we define a beam as inefficient if it serves fewer than F users. After the initial beam–user assignment is completed, all beams are examined sequentially. The users assigned to inefficient beams are released and merged with the pool of unscheduled users. A new round of clustering and beam–user assignment is then performed on the updated user set. The procedure is repeated until either the user capacity requirement is met for all beams, or a maximum number of iterations is reached.

3.2. Algorithm B: Globally Optimized Beam–Slot Assignment Algorithm

In this section, we propose Algorithm B, a globally optimized beam–slot assignment, which allocates the beam footprints obtained from Algorithm A to specific time slots within a BH cycle. The objective is to allocate MN beams across M time slots, where each slot is constrained to activate no more than N beams. To mitigate intra-slot interference and ensure reliable system performance, a minimum spatial separation $d_{\min} = 4R$ must be maintained between any two simultaneously active beams.

The globally optimized beam–slot assignment algorithm combines a greedy initialization with a simulated annealing strategy to maximize the total number of scheduled beams, while satisfying the minimum inter-beam distance constraint and the per-slot capacity limit. The optimization problem is formulated as follows:

$$\max \sum_{t=1}^M |S_t| \quad (10a)$$

$$\text{s.t. } |S_t| \leq N, \forall t \quad (10b)$$

$$\begin{aligned} &\sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \geq d_{\min}, \\ &\forall t, \forall p_i, p_j \in S_t, i \neq j. \end{aligned} \quad (10c)$$

where S_t denotes the set of beams assigned to the t -th time slot, and N represents the maximum number of beams that can be activated per time slot. Each beam is associated with a center coordinate $p_i = (x_i, y_i), i \in \{1, 2, \dots, MN\}$. Constraint (10b) ensures that the

number of beams assigned to any time slot does not exceed N , constraint (10c) enforces that the distance between any two active beams within the same time slot must be greater than a predefined minimum separation $d_{\min}$. We define a state variable $state(b) = t$, $b = \{1, 2, \dots, MN\}$, $t = \{1, 2, \dots, M\}$ to represent that the b -th beam is assigned to the t -th time slot, where $state(b) = -1$ indicates that the b -th beam is not scheduled. The principal parameter settings of the simulated annealing procedure are summarized in Table 1.

Table 1. Simulated annealing optimization simulation parameter settings.

Parameter	Value
Initialize temperature T	500
Minimum temperature $T_{\min}$	10^{-3}
Cooling rate α	0.95
Max iterations $N_{\max}$	500
Acceptance rule	Metropolis criterion $P = e^{-\Delta E/T}$

The pseudocode of Algorithm 1 is as follows:

Algorithm 1 Global Optimization Beam-Time Slot Allocation Algorithm

```

1: Phase 1: Initial Solution Construction Based on Greedy Algorithm Principles
2: Initialize:  $S_t = \emptyset$ ,  $state(b) = -1$ ,  $b = \{1, 2, \dots, MN\}$ ,  $t = \{1, 2, \dots, M\}$ ;
3: for each time slot  $t$  do
4:   for each beam  $b$  do
5:     if  $state(b) = -1$  and  $\forall j \in S_t, \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \geq d_{\min}$  then
6:       Update the beam allocation set  $S_t \leftarrow b$ ;
7:       Update the state variables,  $state(b) = t$ ;
8:     end if
9:     if  $|S_t| = N$  then
10:      break
11:    end if
12:   end for
13: end for
14: Phase 2: Refinement Based on Simulated Annealing Optimization
15: Initialize temperature  $T$ , minimum temperature  $T_{\min}$ , cooling rate  $\alpha$ , max iterations  $N_{\max}$ ;
16: Set the greedy algorithm result as the initial solution  $S_{\text{current}}$ ;
17: while  $T > T_{\min}$  do
18:   for 1 to  $N_{\max}$  do
19:     Randomly select the  $b$ -th beam and remove it from the current allocation  $S_t$ ;
20:     Update beam allocation  $S_t$  and state variables  $state(b) = -1$ ;
21:     for each time slot  $t$  do
22:       if  $|S_t| < N_s$  and  $\forall j \in S_t, \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \geq d_{\min}$  then
23:         Update the beam set  $S'_t \leftarrow b$ ;
24:         Update the state variables,  $state(b) = t$ ;
25:       break
26:     end if
27:   end for
28:   if  $\sum_{t=1}^M |S'_t| > \sum_{t=1}^M |S_t|$  then
29:     Accept the new solution immediately;
30:   else
31:     Accept the new solution with certain probability;
32:   end if
33: end for
34:   Update temperature  $T \leftarrow \alpha T$ .
35: end while

```

3.3. Algorithm C: Joint Frequency–Power Allocation Algorithm

This section presents Algorithm C, which performs the joint optimization of frequency and power resources based on the overall objective defined in objective function P2. For each beam, the total bandwidth is equally divided into $U_{t,b}$ sub-bands, where $U_{t,b}$ is set as equal to the number of users $N_{t,b}$ served by the b -th beam in the t -th time slot, resulting in a sub-band set $\{B_f^{sub} | f = 1, 2, \dots, U_{t,b}\}$. For each user $u_{t,b,f}^i$, the achievable data rate $R_{t,b,f}^i$ across all sub-bands B_f^{sub} is computed and used to construct a reward matrix. The Hungarian algorithm is then applied to solve the resulting assignment problem, aiming to maximize the total system throughput. The frequency allocation problem follows the objective in objective function P2 and is subject to constraints (7f) and (7g).

For power allocation, we first consider intra-beam power distribution among users. Given the beam-specific channel parameters and the total power P_t^b allocated to the b -th beam in the t -th time slot, the intra-beam power allocation can be computed using the classical water-filling algorithm. The objective is to maximize the sum throughput of users within each beam, subject to constraints (7b) and (7c). The channel parameter is denoted as $\lambda_{b,f}^i = \frac{H_{b,f}^i}{N_i B_{b,f}^i}$, where $H_{b,f}^i$ represents the channel gain, including both the transmit gain and path loss. Let μ_b denote the water-filling level corresponding to each beam and $S_{t,b} = \sum_{i=1}^{U_{t,b}} \frac{1}{\lambda_b^i}$. Then, the intra-beam user power allocation result can be obtained as follows:

$$P_{t,b}^i = \mu_b - \frac{1}{\lambda_{b,f}^i} \quad (11)$$

$$\mu_b = \frac{1}{N} (P_t^b + S_{t,b}) \quad (12)$$

The power allocation for users under a single beam of a satellite is given by the following:

$$P_{t,b}^i = \frac{1}{N} (P_t^b + S_{t,b}) - \frac{1}{\lambda_b^i} \quad (13)$$

The inter-beam power allocation is performed based on the results of intra-beam power distribution. At this stage, the throughput of an individual user can be expressed as a function of the beam-level transmit power P_t^b as the following:

$$R_{t,b,f}^i = B_{b,f}^i \log_2 \left[1 + \lambda_{b,f}^i P_{t,b}^i \right] = \log_2 \left[\frac{\lambda_{b,f}^i}{N} (P_t^b + S_{t,b}) \right] \quad (14)$$

The total throughput of all users served by a single beam can then be expressed as follows:

$$\begin{aligned} R_t^b &= \sum_i^{U_{t,b}} B_{b,f}^i R_{t,b}^i = \sum_i^{U_{t,b}} B_{b,f}^i \log_2 \left[\frac{\lambda_{b,f}^i}{N} (P_t^b + S_{t,b}) \right] \\ &= \sum_i^{U_{t,b}} B_{b,f}^i \left[\log_2 \lambda_b^i + \log_2 (P_t^b + S_{t,b}) - \log_2 U_{t,b} \right] \end{aligned} \quad (15)$$

The objective function can then be expressed as follows:

$$\max \sum_t^M \sum_b^N \frac{B_s}{U_{t,b}} \left[U_{t,b} \log_2 (P_t^b + S_{t,b}) + \sum_i^{U_{t,b}} \log_2 \lambda_{b,f}^i - U_{t,b} \log_2 (U_{t,b}) \right] \quad (16)$$

The objective function in (16) aims to maximize the total system throughput and is formulated as a weighted summation of multiple logarithmic terms, where P_t^b is the optimization variable, $S_{t,b}$ is a fixed constant obtained from the preceding water-filling

algorithm, and $\log_2 \lambda_{b,f}^i$ and $U_{t,b} \log_2(U_{t,b})$ are constant terms independent of the optimization variable. Since $\log_2(P_t^b + S_{t,b})$ is a strictly concave function over its domain and both $P_t^b > 0$ and $S_{t,b} > 0$ hold, the objective remains concave. Moreover, the constraints of the optimization problem (7d) and (7e) are linear and do not affect the concavity of the objective. The form of this problem is consistent with the power allocation formulation in [23], and can therefore be efficiently solved using standard convex optimization methods.

4. Simulation Results and Analysis

4.1. Simulation Parameters

In the simulation scenario considered in this paper, a circular ground area with a radius of 700 km is selected as the target service region. This area is jointly covered by four LEO satellites and one GEO satellite, with each LEO satellite operating in a multi-beam mode and capable of activating up to four beams simultaneously. Over six time slots, the LEO satellite system can form up to 96 beam positions that can be scheduled under this configuration. In each time slot, the GEO satellite activates one beam and establishes a protection zone with a radius of 150 km around GEO user. Beam coverage from the LEO satellites is prohibited within this protection zone to avoid interference. The simulation parameters are summarized in Table 2.

Table 2. Simulation parameter settings.

Parameter	Value
LEO Satellite Orbital Altitude	500 km/600 km
Number of LEO Satellites	4
GEO Satellite Position	40° E, 100° N
GEO Earth Station Location	40° E, 100° N
Radius of Target Ground Area	700 km
LEO Satellite Beam Radius	50 km
Peak Antenna Gain of LEO Satellite	35 dB
3 dB Beamwidth	1.65°
Number of Beams per LEO Satellite	4
Number of Users to Be Served	750
Number of Time Slots	6
System Bandwidth	400 MHz
Carrier Frequency	20 GHz
Noise Temperature	150 K
Maximum LEO Satellite Transmit Power	800 W
Rician Fading Factor	0.95
Cloud/Rain Attenuation Coefficient	0.1/0.05
Maximum Beam Power	250 W

The antenna gain used in the simulation is calculated according to the following formula [28,29]:

$$G_b^i = G_0 \left[\frac{J_1(\mu)}{2\mu} + 36 \frac{J_3(\mu)}{\mu^3} \right]^2 \quad (17)$$

where G_0 denotes the peak transmit antenna gain of the b -th beam, which is calculated as follows: $G_0 = \eta \frac{4\pi D}{(c/f)^2}$, where η is the antenna aperture efficiency, D is the antenna aperture diameter, f is the carrier frequency, and c is the speed of light. Then $\mu = 2.07123 \frac{\sin \Theta_b}{\sin \Theta_{3dB}}$, where Θ_{3dB} denotes the 3 dB beamwidth of the b -th beam, and Θ_b represents the off-axis angle between the i -th user and the pointing direction of the b -th beam. $J_1(\cdot)$ and $J_3(\cdot)$ are the first- and third-order Bessel functions of the first kind, respectively.

4.2. Performance of the Beam-Footprint Partitioning Algorithm

In this section, we evaluate the performance of the beam-footprint partitioning algorithm. The beam positions generated by Algorithm A are assigned to time slots using Algorithm 1. If Algorithm A produces only MN beam positions, the assignment performance of Algorithm 1 may degrade in certain scenarios due to the spatial separation constraints among beams within each time slot. This can lead to a reduced number of active beams scheduled within a BH cycle, thereby lowering overall resource utilization. To address this issue, an adjustment factor ∂ is introduced. Specifically, Algorithm A generates ∂MN candidate beams, and Algorithm 1 selects MN beams from this larger pool for actual scheduling. In the simulation, $\partial = 1.5$. Each LEO satellite can activate up to 4 beams simultaneously, resulting in a maximum of 24 beam positions that can be scheduled over six time slots. Based on this configuration, a total of 144 candidate beams are generated across four LEO satellites. Theoretically, the maximum number of users that can be scheduled is 432.

Without the iterative optimization mechanism, a significant number of beams serve fewer than F users, resulting in a total of 408 users being scheduled, with a resource utilization rate of 94.4%, as illustrated in Figure 2a. After introducing iterative refinement, the number of inefficient beams is significantly reduced. Consequently, the number of scheduled users increases to 430, and the resource utilization rate improves to 99.5%, as shown in Figure 2b.

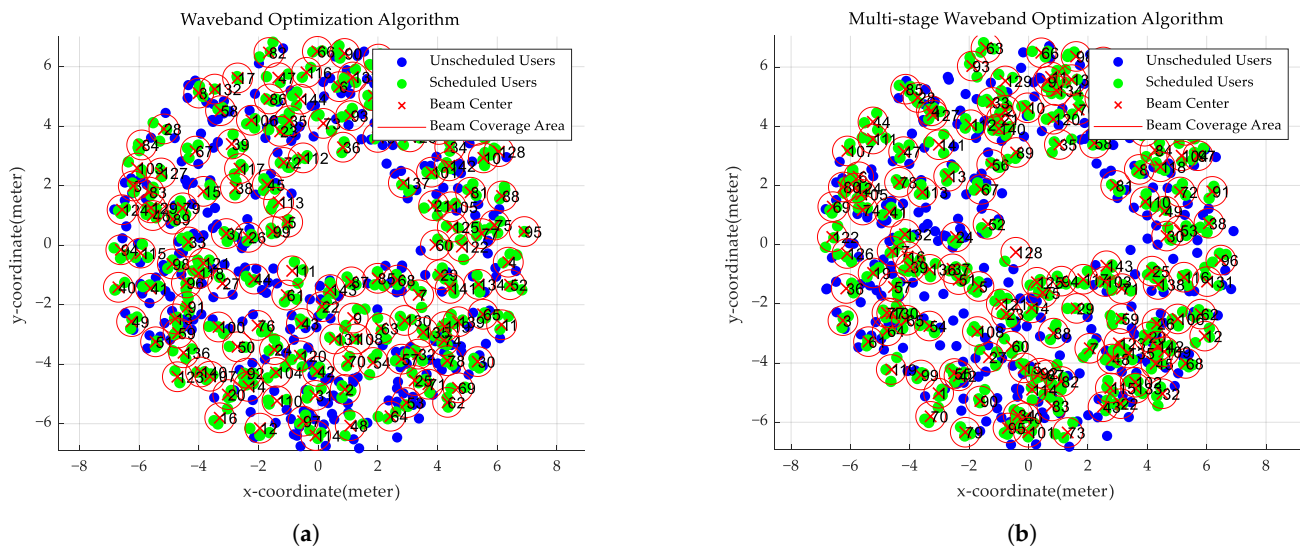


Figure 2. Performance of beam-footprint partitioning algorithm. (a) Beam-slot assignment without iterative optimization; (b) beam-slot assignment with iterative optimization.

4.3. Performance of the Globally Optimized Beam-Slot Assignment Algorithm

In this section, we evaluate the performance of the globally optimized beam-slot assignment algorithm by comparing the beam distributions across time slots before and after applying simulated annealing. Under the simulation settings described in Section 4.1, the maximum number of beams that can be activated over six time slots is 96. Without simulated annealing, the actual number of activated beams is 90, as shown in Figure 3a. After applying the annealing-based iterative optimization, the number increases to 96, as illustrated in Figure 3b.

The comparison reveals that, without simulated annealing, beam allocation in the first five time slots is based purely on a greedy strategy without global optimization. As a result, the sixth time slot suffers from a lack of available beam positions, significantly reducing the total number of active beams during the BH cycle and degrading overall

system performance. In contrast, the inclusion of simulated annealing enables global optimization across all time slots. By introducing random perturbations, the algorithm escapes local optima that may arise from greedy allocation strategies, thereby increasing the number of activated beams from 90 to 96, which corresponds to the theoretical upper bound of system performance.

The proposed hybrid scheduling algorithm, combining greedy initialization with simulated annealing, exhibits strong scalability. Its computational complexity grows polynomially with the number of candidate beams MN and is largely insensitive to variations in user density. Experimental results under a large-scale scenario (10 satellites, 240 beams, and 2500 users) show that the algorithm completes in 0.015 s while activating all 240 beams, achieving near-global optimality and effectively avoiding the local optima commonly encountered by purely greedy strategies. Moreover, the simulated annealing phase can be further accelerated through multi-threaded perturbation, enabling the proposed method to scale efficiently to even larger LEO constellation systems.

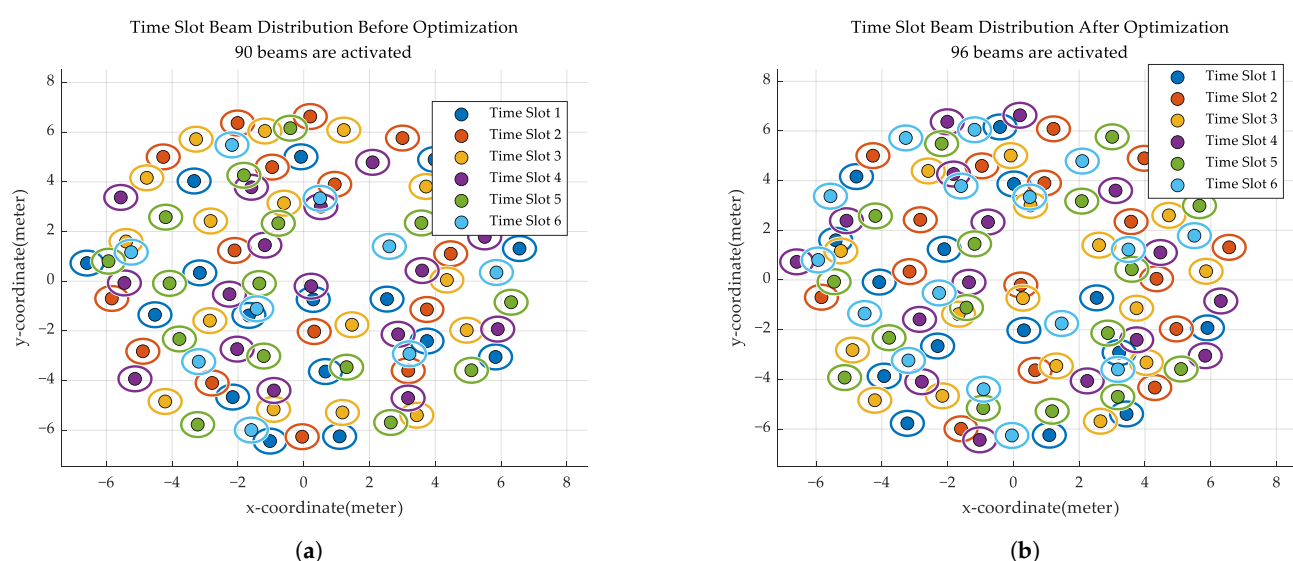


Figure 3. Performance of globally optimized beam–slot assignment algorithm. (a) Beam–slot distribution before optimization; (b) beam–slot distribution after optimization.

4.4. Performance of the Joint Frequency–Power Allocation Algorithm

In this section, we evaluate the performance of the proposed joint frequency–power optimization algorithm through simulation. For power allocation, three baseline schemes are used for comparison: (1) uniform power allocation across both beams and users, (2) uniform power allocation across beams only, and (3) uniform power allocation across users only. The simulation results comparing all four schemes are presented in Figure 4.

The simulation results indicate that optimizing power allocation either across beams or within beams can significantly enhance system throughput. Among the four evaluated schemes, Algorithm C, which performs joint power allocation for both satellite beams and users, achieves the highest performance gain. It improves the total system throughput by approximately 7.2% to 20% compared to the other uniform power allocation strategies.

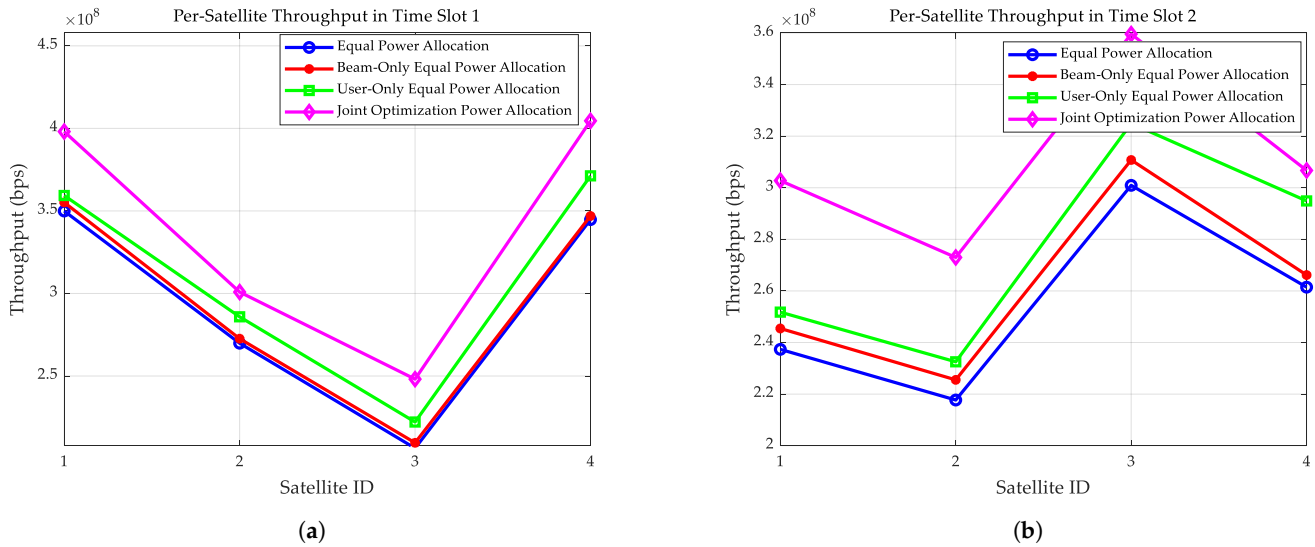


Figure 4. Performance of joint frequency–power allocation algorithm. (a) Throughput comparison of different power allocation algorithms for each satellite in time slot 1; (b) throughput comparison of different power allocation algorithms for each satellite in time slot 2.

4.5. Computational Complexity and Runtime Evaluation

We analyze the computational complexity of each sub-algorithm as follows.

For Algorithm A, in the beam generation stage, K-means clustering is applied. Each iteration requires computing the distances between U users and MN beam centers. If the number of iterations is denoted by N_k , the complexity of this process is $O(N_k UMN)$. In beam–user matching, an ILP formulation is adopted. Based on empirical results from commercial solvers, the complexity of the ILP grows approximately with the number of variables to the power of 1.3 to 1.6. Therefore, the complexity can be estimated as $O((UMN)^{1.5})$. Hence, the total complexity of Algorithm A is $O(N_k UMN + (UMN)^{1.5})$. This complexity scales polynomially with both the number of users U and the number of candidate beams MN , making it sensitive to large user populations. However, the modular structure ensures that beam generation and assignment are handled efficiently and converge quickly in practice. Therefore, even for large-scale LEO constellations, Algorithm A remains computationally feasible.

For Algorithm 1, during greedy initialization, each candidate beam must be evaluated for placement within a time slot by checking the minimum distance constraint with up to N existing beams. This results in a complexity of $O(MN^2)$. In the simulated annealing phase, assume the total number of random perturbations is N_{SA} . Each perturbation involves randomly selecting a beam and checking distances against up to N other beams in the new slot, yielding a complexity of $O(N_{SA}N)$. Therefore, the overall complexity of Algorithm 1 is $O(MN^2) + O(N_{SA}N) = O(MN^2)$. Since Algorithm 1 only depends on the number of beams and time slots, but not on the number of users U , it exhibits excellent scalability in scenarios with dense user deployments. Its complexity increases quadratically with the number of beams per slot but remains tractable for typical satellite system parameters. This makes it suitable for real-time or near-real-time scheduling in large constellations.

For Algorithm C, this algorithm involves sub-band assignment, intra-beam power allocation, and inter-beam power coordination. Sub-band assignment is performed using the Hungarian algorithm. For each beam, a reward matrix of dimension $L \times L$ is constructed, where L is the maximum number of users per beam. The complexity of Hungarian matching per beam is $O(L^3)$, and the total complexity is $O(MNL^3)$. Intra-beam power allocation using the water-filling method requires ordering the channel gain values. The per-beam

complexity is $O(L \log L)$, thus a total complexity of $O(MNL \log L)$. Inter-beam power control is formulated as a convex optimization problem, resulting in a total of $O(MN)$. Therefore, the total complexity of Algorithm C is $O(MNL^3) + O(MNL \log L) + O(MN)$. The complexity of Algorithm C grows with the number of beams MN and the number of users per beam L , but remains polynomial overall. This ensures tractability even in high-throughput systems where each beam serves many users. The use of efficient algorithms like Hungarian matching and water-filling further improves practicality, making this approach well-suited for dynamic resource management in large-scale satellite systems.

The experiments were conducted under a hardware environment consisting of a Windows 11 operating system, an Intel i9-13900H processor (Intel Corporation, Santa Clara, CA, USA), and 16 GB of RAM. Based on the simulation settings described in Section 4.1, the average runtime of each algorithm is as follows: Algorithm A requires approximately 1.76 s, Algorithm 1 requires approximately 0.003 s, and Algorithm C requires approximately 5 s.

5. Conclusions

This paper proposes a beam-hopping-based joint beam scheduling and resource allocation framework, which leverages beam hopping technology to perform global scheduling of beam positions across time slots, enabling efficient multi-dimensional optimization in time, space, frequency, and power domains. The overall resource allocation problem is decomposed into three subproblems: beam-footprint partitioning, beam-to-slot assignment, and joint frequency–power allocation.

First, in the beam partitioning stage, a multi-stage optimization algorithm is introduced to maximize the number of users served by each beam through ILP-based modeling and iterative refinement. Second, for beam-to-slot assignment, a combination of greedy and simulated annealing algorithms is employed to optimize beam allocation across time slots, aiming to reduce inter-beam interference and improve resource utilization efficiency. Finally, in the frequency and power allocation stage, the Hungarian matching algorithm and water-filling method are used to efficiently allocate sub-bands and beam power, with the objective of maximizing user throughput.

Simulation results demonstrate that the proposed dimension-wise optimization approach significantly improves resource utilization and throughput with relatively low computational complexity. Compared to traditional joint iterative optimization methods, the proposed method avoids the heavy computational burden caused by multi-dimensional iterative coupling. Under the simulation settings used in this study, the system achieves a resource utilization rate of 99.5%, with a total user throughput improvement ranging from 7.2% to 20%.

The proposed framework provides a flexible and scalable solution for dynamic resource management in LEO satellite systems. By decoupling the complex joint optimization into well-structured subproblems and integrating both exact and heuristic algorithms, it significantly enhances system performance in terms of throughput and resource efficiency. Importantly, this framework also offers a promising direction for addressing the coexistence challenges in heterogeneous satellite networks, such as LEO–GEO integration, by enabling coordinated and interference-aware scheduling across multiple dimensions. As such, it lays a solid foundation for the development of intelligent, adaptive, and high-capacity satellite systems in future space–air–ground integrated networks.

However, the current scheduling framework is primarily based on offline computation and assumes static system states within each BH cycle. In highly dynamic orbital environments or under burst-like traffic demands, the responsiveness and timeliness of scheduling decisions become critical. Future research could incorporate real-time scheduling mechanisms that address computational latency and rescheduling strategies in response

to dynamic user access and satellite mobility. For example, task priority and link-state awareness could be integrated to design fast-response dynamic scheduling strategies suited for more complex operational scenarios.

While this paper has demonstrated the effectiveness of the proposed method in medium-scale systems, larger-scale satellite constellations and higher user densities will significantly increase resource coupling and optimization dimensions. To this end, future work may consider developing distributed or hierarchical scheduling frameworks tailored for ultra-large-scale constellations—for instance, by enabling inter-layer coordination across different orbits to achieve globally efficient resource allocation. In addition, incorporating intelligent methods such as reinforcement learning or graph neural networks to enable context-aware scheduling strategies presents a promising direction for further exploration.

Author Contributions: Conceptualization, J.Z., W.L. and Y.L.; Methodology, J.Z., W.L. and S.L.; Software, J.Z. and H.W.; Writing—original draft, J.Z.; Writing—review and editing, H.W., Y.L. and J.Z.; Funding acquisition, W.L. and S.L. All authors have read and agreed to the published version of the manuscript.

Funding: This work was supported in part by the Beijing Natural Science Foundation under grant L222003, and in part by the National Key R&D Program of China under grant 2024YFB2907600.

Data Availability Statement: Data is contained within the article.

Acknowledgments: The authors would like to thank all reviewers for their helpful comments and suggestions regarding this paper.

Conflicts of Interest: The authors declare no conflicts of interest.

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